Network Uncertainty at the Resolution of the Data

Observation Grain, Reporting Clock, and Regularization
in Agent-Based SIR Models

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Abstract

Pandemic transmission depends on contact network structure. The mean-field ODE SIR model is therefore strictly misspecified as a mechanistic account: the SI/N term encodes uniform mixing, not actual contact topology. But misspecification alone does does not determine how uncertain inference about the network mechanism should be. The central question is how the observation process—the grain at which infections are measured and the clock at which they are reported—changes the uncertainty that remains over network structure.

We treat the network as an uncertain object and compare posterior uncertainty under contact-network, group-level incidence, and aggregate incidence observations. Direct contact observation sharply constrains community structure; incidence-only schemes leave substantial uncertainty, with group grain more informative than aggregate grain and finer aggregate timing insufficient by itself. Half-normal, exponential, and spike-and-slab priors over the community parameter expose the central tradeoff: shrinkage toward uniform networks prevents unsupported structural claims but accepts conservative bias when observations cannot justify heterogeneity. As a cautionary reduction, we also calibrate a mean-field ODE to weekly aggregate observations. Its posterior concentrates near a pseudo-true parameter that absorbs network structure into a lower effective aggregate rate, understating the density-matched mean-field reference by 8.5% in our baseline experiment. Collecting more independent outbreaks at the same grain and clock reduces posterior width fourfold without reducing bias. More data make the wrong model more certain when the additional data are collected under the same insufficient observation process. We connect this to PAC-Bayesian regularization: the ABM supplies a distribution over heterogeneous networks, while grain and clock determine how far observation can move that distribution away from its regularizing prior.

1 Calibration Requires an Observation Model

ABMs generate rich latent trajectories, but calibration data are usually coarser [Epstein, 2008]. We therefore treat an ABM as a generative statistical model:

$$\text{simulator} \;\rightarrow\; \text{latent social process} \;\rightarrow\; \text{observation process} \;\rightarrow\; \text{data.}$$

The simulator encodes interaction; the observation process encodes what is measured. *Grain* is the level observed—edges, individual events, groups, or population totals. *Clock* is the reporting interval—event time, daily, weekly, or monthly. These define different calibration problems, not merely datasets of different quality. Our inferential target is therefore a distribution over network parameters conditional on a specified grain and clock.

2 Mean-Field and Network SIR

The standard mean-field ODE SIR model [Kermack and McKendrick, 1927, Keeling and Rohani, 2008] is:

$$\frac{dS}{dt} = -\beta_{ODE}\frac{SI}{N}, \qquad \frac{dI}{dt} = \beta_{ODE}\frac{SI}{N} - \gamma I, \qquad \frac{dR}{dt} = \gamma I.$$

The product SI/N encodes uniform mixing: it is a social assumption, not merely algebra. A network SIR instead gives susceptible agent i infection probability [Keeling and Rohani, 2008, Kiss et al., 2017]

$$P\big(i \text{ infected in } [t,\, t+\Delta t]\big) = 1 - \exp\Big(-\beta\,\Delta t \sum_{j\in\mathcal{N}(i)} \mathbf{1}\{X_j(t) = I\}\Big).$$

so aggregate incidence depends on susceptible–infected edges [Newman, 2002]. The per-contact β is not β_{ODE}. Under uniform mixing with mean degree $\bar{d}$, their density-matched benchmark is

$$\beta_{mf,true} = \beta \cdot \bar{d}.$$

This is a benchmark, not microscopic truth. Community structure breaks the approximation between edge counts and SI, producing group-wise epidemic waves that a constant-rate ODE absorbs into a pseudo-true effective parameter.

3 Regularization and the Uniform Graph

We generate contact networks from a stochastic block model [Holland et al., 1983]:

$$\operatorname{logit} P(A_{ij} = 1) = \alpha + \eta\, B_{ij},$$

where $B_{ij} = 1$ if agents i, j share a group and $B_{ij} = -1/(K-1)$ otherwise. The parameter η controls community structure; α controls overall density. Because the logistic link is nonlinear, mean-zero coding of B_{ij} does not by itself preserve density. We therefore solve for $\alpha(\eta)$ at each value of η so that expected density is held fixed. We place the prior on the graph-generating parameter, not on a single realized adjacency matrix:

$$\eta \sim P, \qquad A \mid \eta \sim \mathrm{SBM}(\alpha(\eta), \eta).$$

This hierarchy induces a distribution over networks while keeping inference computationally tractable and preserving graph-realization uncertainty conditional on η.

At $\eta = 0$ the graph is Erdős–Rényi: uniformity is the conservative baseline, not a substantive truth. We compare half-normal, exponential, and spike-and-slab priors at three strengths. All favor $\eta = 0$; the spike-and-slab also assigns exact uniformity positive mass. Regularization preserves the ABM's capacity for heterogeneity while requiring observations to justify using it.

4 Mean-Field Collapse as a Cautionary Example

Setup. We simulate $N = 1000$ agents in four groups with $\eta = 2.5$, mean degree 12.2, and within-group edge share 0.90. We set per-contact $\beta = 0.04$ and $\gamma = 0.10$, observe 150 days as weekly aggregate incidence, and fit β_{ODE} with γ fixed using

$$\log(1 + Y_k) \sim \mathcal{N}\big(\log(1 + \hat{Y}_k),\, \sigma^2\big).$$

The ODE posterior concentrates near $\beta^*_{ODE} \approx 0.447$, below the density-matched benchmark $\beta_{mf,true} \approx 0.489$, an 8.5% bias. It absorbs unobserved community waves into a lower effective rate.

More data, same grain. We generate R independent community-network outbreaks and pool their likelihoods. As R increases from 1 to 20, the posterior 95% credible interval width shrinks from 0.0268 to 0.0071—a nearly fourfold reduction. The posterior MAP stays in the range [0.358, 0.396], always below $\beta_{mf,true} \approx 0.489$.

5 Network Uncertainty and PAC-Bayesian Regularization

We quantify uncertainty in η under each observation process and prior. For incidence observations, we estimate a simulation-based synthetic likelihood from 20 replicated epidemics per grid point and evaluate it on 30 independent held-out epidemics generated under both a uniform network ($\eta = 0$) and a structured network ($\eta = 2.5$). The prior comparison makes the intended tradeoff explicit. Under the uniform generator, a moderate spike-and-slab prior leaves less posterior

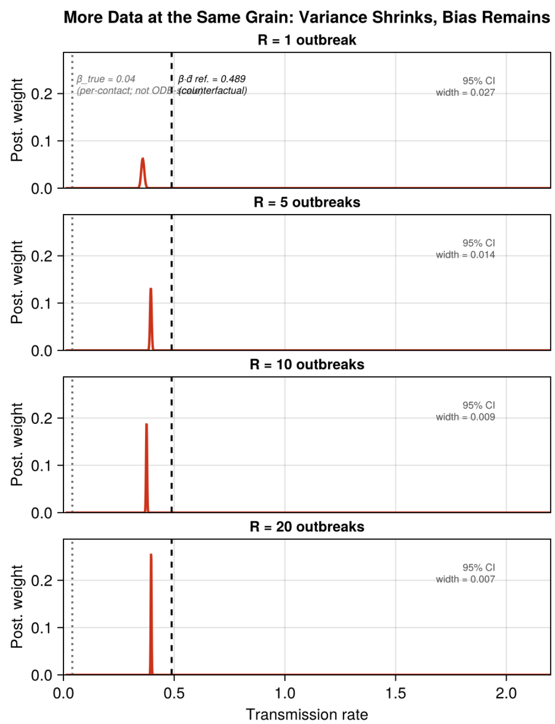

Figure 1: Pooling $R = 1, 5, 10, 20$ weekly aggregate outbreaks narrows the ODE posterior nearly fourfold without moving it to the density-matched reference (dashed). Repetition at the same grain and clock sharpens the reduced model.

mass on spurious structure than a moderate half-normal prior; for group-weekly data, the corresponding mass above $\eta = 0.25$ is 0.27 versus 0.58. Under the structured generator, the same conservative prior increases downward bias: the mean group-weekly posterior for η is 1.32 under spike-and-slab and 1.50 under half-normal. Aggregate observations are shrunk still more strongly. This is the desired behavior: when grain and clock provide weak evidence about topology, we accept conservative bias rather than confidently hallucinating structure.

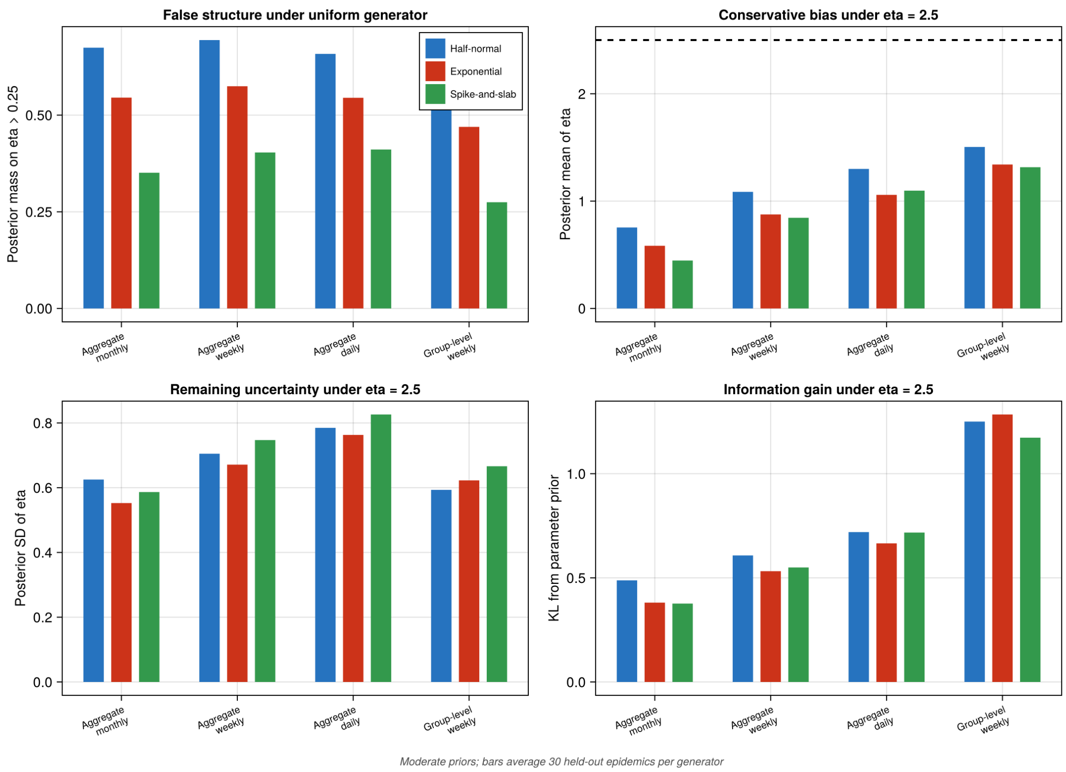

Figure 2: Prior sensitivity and observation-conditioned network uncertainty for moderate half-normal, exponential, and spike-and-slab priors. Top left: false structural mass under a uniform generator. Top right: shrinkage of the posterior mean under a structured generator ($\eta = 2.5$, dashed reference). Bottom: posterior spread and information gain under the structured generator. Stronger concentration near uniformity suppresses invented structure at the cost of conservative bias when structure is present. Bars average 30 held-out epidemics.

In a PAC-Bayesian formulation [McAllester, 1999], candidate network structures or network-generating parameters are evaluated by predictive performance penalized by departure from a prior over simpler or more regular mechanisms. Under rich observation—such as contact edges or exposure histories—the posterior can move substantially away from the regularizing prior. Under coarse observation—such as weekly aggregate incidence—many network configurations can retain posterior mass. For observation operator $\mathcal{O}_{g,c}$ and its associated calibration loss, the generalized posterior takes the form

$$Q_{g,c}(d\eta) \propto \exp\{-\tau \widehat{L}_{g,c}(\eta)\}\, P(d\eta), \qquad A \mid \eta \sim P(A \mid \eta).$$

Equivalently, $Q_{g,c}$ trades posterior-averaged predictive loss against $\mathrm{KL}(Q_{g,c} \| P)$. Grain and clock therefore enter the loss through the observation operator, while the parameter prior determines how unsupported network structure is regularized.

PAC-Bayes offers a framework for quantifying which network distinctions the data support and how uncertain unsupported distinctions should remain. This is the appropriate inferential logic for agent-based models: retain heterogeneity when the data support it, penalize it when they do not, and quantify uncertainty when the observation process cannot decide. A mechanism cannot be validated at a resolution where its identifying information has been discarded. More data help only when they add information relevant to the mechanism, not merely more replications of the same reduced observation.

This framing clarifies the role of priors. A prior is a form of regularization: it encodes which model configurations are penalized in the absence of data support, exactly as a penalty term shrinks parameters toward a simpler structure [Tibshirani, 1996]. Priors and perturbation schemes are in this sense dual—both express assumptions about which departures from a baseline model should require justification from the data [Hazan et al., 2016]. The practical consequence is that disputes over which prior is *correct* are largely unproductive. No prior is uniquely correct; each encodes a different regularization. What is productive is comparing priors: if conclusions are stable across a range of reasonable priors, they are robust to that modeling choice. If they are not, the instability is itself informative about what the data can and cannot identify [Gelman and Shalizi, 2013].

A practical workflow follows: (1) specify the latent process and observation process separately; (2) place a regularizing prior over network structure; (3) condition that distribution under alternative grains and clocks; (4) compare posterior entropy, spread, and information gain; (5) propagate network uncertainty to epidemic and intervention predictions; (6) run posterior predictive checks.

6 Conclusion

Mean-field SIR is not wrong because it is mathematically crude; it is wrong because it encodes a social theory of uniform contact. Whether that wrongness matters empirically depends on the observation process. The inferential target should be a distribution over compatible networks whose uncertainty is explicitly conditional on observational grain and clock. Regularized network inference retains heterogeneity when observations constrain it and preserves uncertainty when they do not.

Under weekly aggregate incidence, additional outbreaks reduce posterior variance around a pseudo-true ODE parameter but do not recover the network mechanism. More data at the same grain and clock estimate the wrong reduced model more precisely. The fully qualified claim is this: more data make the wrong model more certain when the additional data are collected under the same insufficient observation process. Direct contact-network observation sharply reduces uncertainty about community structure in our experiment. Group-level incidence produces a smaller but meaningful update, while finer aggregate timing alone yields limited information about topology. The point is not to declare one network recovered or unrecovered, but to quantify the distribution of networks compatible with each observation process. Coarse data require regularized uncertainty rather than false mechanistic confidence.

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